Supplement

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Scopus search term: (TITLE-ABS-KEY ("pseudo*" OR "jackknife*" OR "leave-one-out" OR (leave W/5 out))) AND (TITLE-ABS-KEY ((time W/5 analy*) OR "survival" OR "censor*")) AND (ALL ("simulat*")) AND (SRCTITLE ("BMC MEDICAL RESEARCH" OR "BIOMETRIKA" OR "BIOSTATISTICS" OR "STATISTICAL METHODS IN MEDICAL RESEARCH" OR "BIOMETRICAL JOURNAL" OR "STATISTICS IN MEDICINE" OR "BIOMETRICS" OR "INTERNATIONAL JOURNAL OF BIOSTATISTICS" OR "COMPUTATIONAL STATISTICS" OR "JOURNAL OF THE AMERICAN STATISTICAL ASSOCIATION")) AND PUBYEAR > 2002 AND PUBYEAR < 2026

Data-Generating Process

The effect of the treatment allocation variable A on Y is confounded with X because the covariate distributions in the treatment groups are different, like in an observational study.

Figure **??** shows the causal structure of the correlated covariates X_1, X_2, X_3, X_4, treatment allocation variable A, and survival time T. X_1 influences both A and T, X_2 only A, X_3 has no direct effect on either A or T but is highly correlated with the other covariates, and finally, X_4 influences only T.

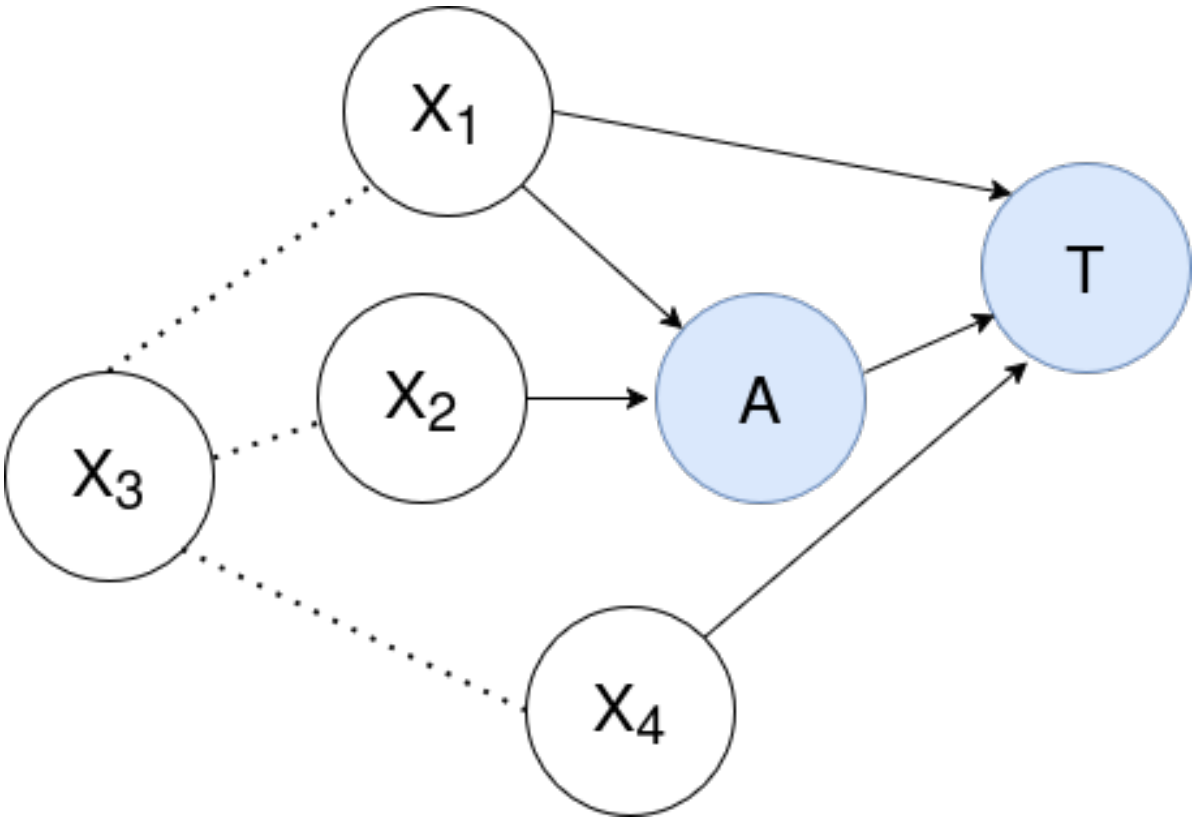


Figure 1: A graph visualizing the relationship between the variables X_1, X_2, X_3, X_4, treatment allocation A and survival time T.

Let $X = \begin{pmatrix} X_1 \\ X_2 \\ X_3 \\ X_4 \end{pmatrix} \sim N_4(0, \Sigma)$, where each individual variance corresponding to the diagonal elements of Σ is 1 and the covariances are given by these pairwise correlations:

$$\begin{aligned} &\text{cor}(X_1, X_2) = 0.1, && \text{cor}(X_1, X_3) = 0.5, && \text{cor}(X_1, X_4) = 0.1, \\ &\text{cor}(X_2, X_3) = 0.6, && \text{cor}(X_2, X_4) = -0.2, && \text{cor}(X_3, X_4) = -0.5. \end{aligned}$$

The treatment assignment is denoted as $A \in \{0, 1\}$ and had the following probability:

$$\mathbb{P}(A = 1 | X_1, X_2) = logit^{-1}(-0.5 + 0.5X_1 + 0.8X_2) \tag{1}$$

This means that higher values of X_1 and X_2 correspond to a higher probability of receiving the treatment. This becomes apparent in Figure 2, where the treatment group has higher median

values for X_1, X_2 and also X_3 because of it's correlation structure. Since X_4 is negatively correlated with X_2, X_3 and only has a slightly positive correlation with X_1, the median of X_4 in the treatment group is slightly lower than in the control group.

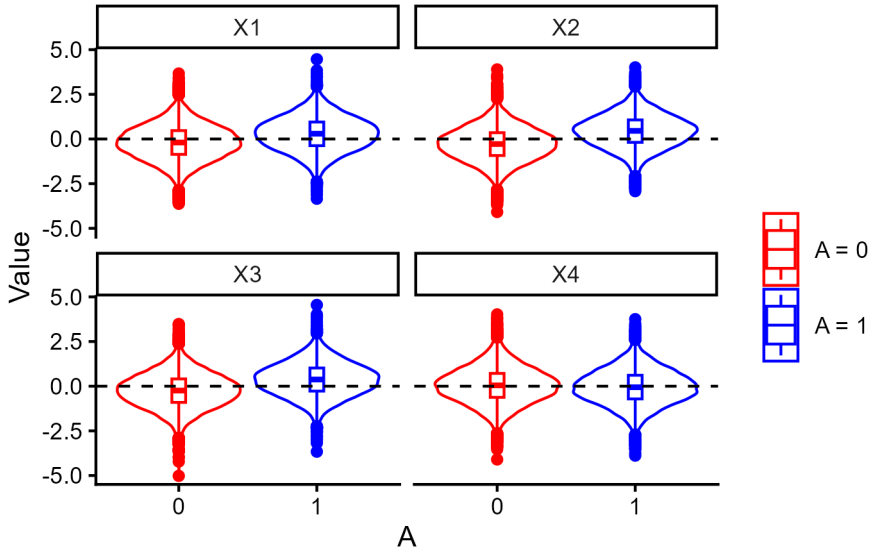


Figure 2: A plot showing the distribution of X_1, X_2, X_3, X_4 between control and treatment groups.

Survival times were generated from a Weibull distribution with:

- shape parameter: $k_i = 1 + \gamma \cdot A_i$, where $\gamma = 0$ corresponds to proportional hazards.
- scale parameter: $\lambda_i = \exp(\beta_1 X_{i1} + \beta_4 X_{i4} + \beta_5 \cdot A_i)$,

Then survival times are given by:

$$T_i = \left(\frac{-\log(U_i)}{\lambda_0 \cdot \lambda_i} \right)^{1/k_i}, \quad U_i \sim \text{Uniform}(0, 1) \tag{2}$$

where $\lambda_0 = 0.05$ is a constant multiplied with the scale parameter λ_i.

The censoring time was drawn from a mixture distribution:

$$C_i = 0.7 \cdot U_i + 0.3 \cdot \min(E_i, 30) \tag{3}$$

where $U_i \sim \text{Uniform}(0, c_{\max})$, $E_i \sim \text{Exponential}(\lambda_C)$. Taking $0.3 \cdot min(E_i, 30)$ ensures that no value greater than 9 can be contributed from the exponential distribution. The distribution

of censoring times is found further down in Figure 4. It should be noted that this form of censoring is *entirely independent* of covariate values.

Observed time and event indicator (i.e. status) are defined as

$$\tilde{T}_i = \min(T_i, C_i), \quad \delta_i = \mathbb{I}(T_i \leq C_i)$$

which produced a dataset that contains:

$$\{\tilde{T}_i, \delta_i, X_{i1}, X_{i2}, X_{i3}, X_{i4}, A_i\}_{i=1}^n$$

Scenarios

Based on the covariate adjusted difference in $RMST(\tau = 10)$ between the control and treatment group (i.e. the ATE for $RMST(\tau = 10)$), *four populations* with *three different amounts of censoring* were simulated, for a total of *12 scenarios*.

The four populations were simulated by four pairwise parameter sets: for each population $p \in \{1, 2, 3, 4\}$, the parameters were assigned as $\beta_5 = \{-0.5, 0, 0.5, -1\}_p$ and $\gamma = \{0, 0, -0.3, 0.49\}_p$. That is, the p-th element from each vector was used together.

Population 1 shows a clear positive ATE (ATE=0.777, i.e. group with $A = 1$ has higher RMST) with proportional hazards. Population 2 has the exact same survival curves with proportional hazards and thus no ATE at all (ATE=0). Population 3 corresponds to no difference in the ATE (ATE=0, raw ATE = 0.399) even though the survival curves are crossing (=non-proportional hazards). Population 4 shows a mild positive ATE (ATE=0.224, raw ATE = 0.528) with non-proportional hazards and crossing survival curves. Descriptive and True were calculated as written in the ATE chapter in the main paper.

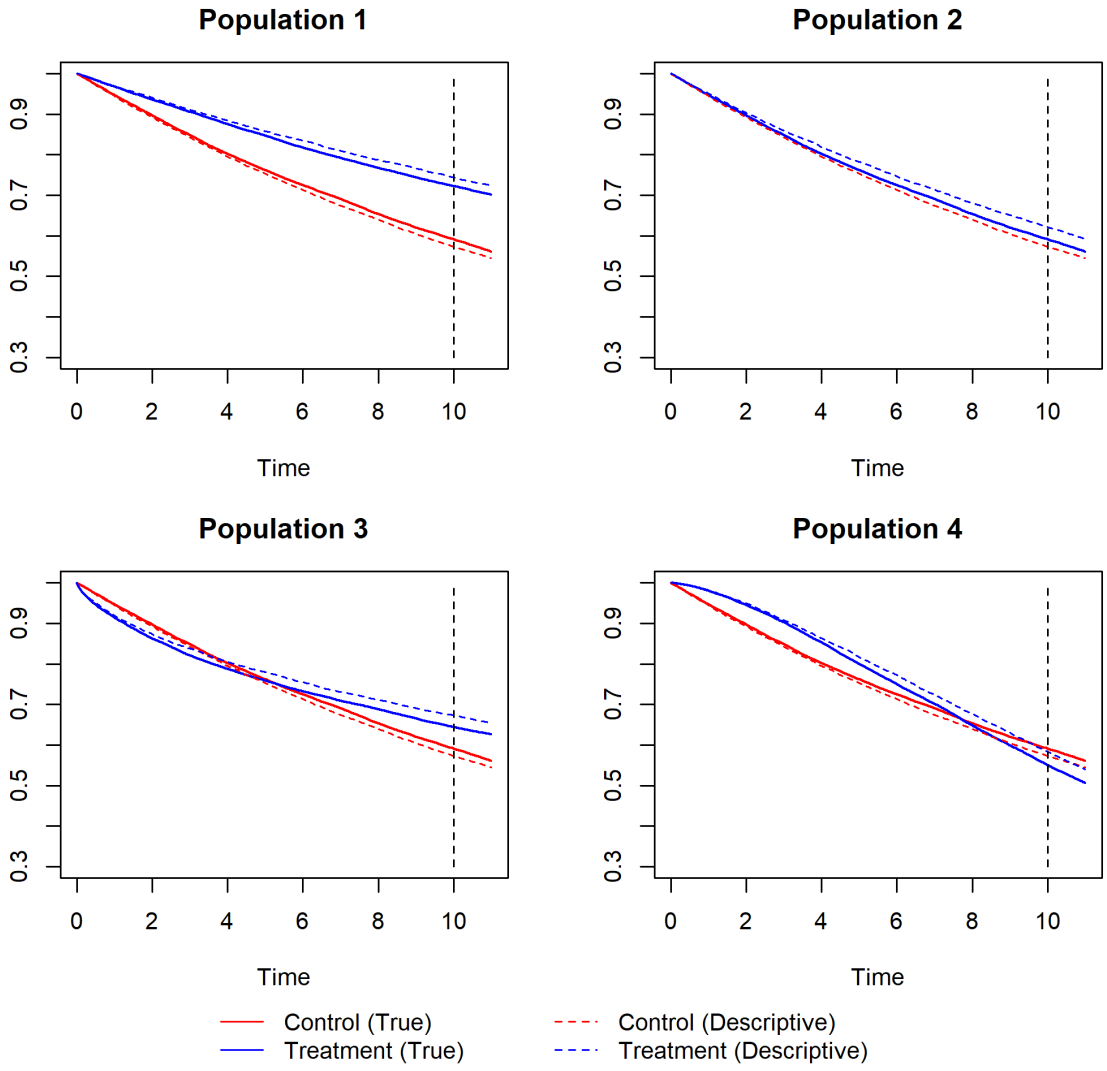


Figure 3: Survival functions ("True") and Kaplan-Meier estimated survival curves ("Descriptive", unadjusted for X) of the treatment groups of all populations.

20% censoring was reached by setting $c_{max} = 20$ and $\lambda_C = 0.02$. 40% was reached with $c_{max} = 20$ and $\lambda_C = 0.06$. Finally, 60% censoring was reached by $c_{max} = 15$ and $\lambda_C = 0.08$. Figure 4 shows the cumulative distribution of censoring times when these values are inserted into the mixture distribution of 3.

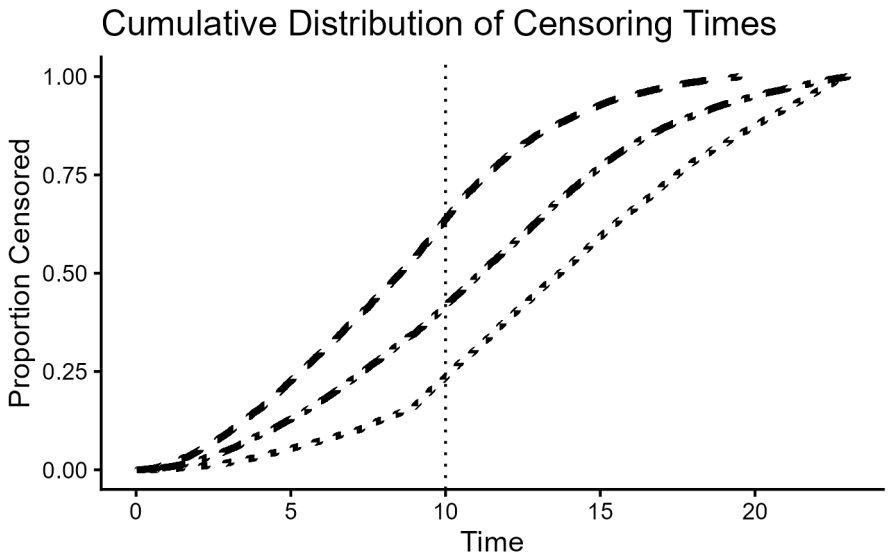


Figure 4: Cumulative Distribution of censoring times, reaching 20%, 40% and 60% censoring at $\tau = 10$.